INCH-POUND

MIL-F-47249A (MI)
29 June 1992
SUPERSEDING
MIL-F-47249 (MI)
26 July 1974

MILITARY SPECIFICATION

FILM, POLYAMIDE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of polyamide film.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8135

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-2073-1	-	DOD Materiel Procedures for Development and Application of Packaging Requirements (Part 1 of 2 Parts)
MIL-STD-2073-2	-	Packaging Requirement Codes (Part 2 of 2 Parts)

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 882	-	Tensile Properties of Thin Plastic Sheeting, Standard Test Methods for
ASTM D 1004	-	Initial Tear Resistance of Plastic Film and Sheeting, Standard Test Method for
ASTM D 1204	-	Linear Dimensional Changes of Nonrigid Thermoplastic Sheeting or Film at Elevated Temperature, Standard Test Method for
ASTM D 1505	-	Density of Plastics by the Density-Gradient Technique, Standard Test Method for

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ASTM D 1790	-	Brittleness Temperature of Plastic Sheeting by Impact, Standard Test Method for
ASTM D 1922	-	Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method, Standard Test Method for
ASTM E 96	-	Water Vapor Transmission of Materials, Standard Test Methods for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material.

3.2.1 Composition. The material shall be a nylon-6-type thermoplastic film made from a polymer of caprolactam, treated on both sides for bondability.

3.2.2 Thickness. Film thickness shall be 0.001 ± 0.0002 inch (see 4.7.9).

3.3 Performance characteristics. The finished film shall conform to the requirements specified in table I.

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TABLE I. Performance characteristics.

Property	Requirements
Specific gravity	1.13 ± 0.02
Tensile strength MD XD	62.053 lbf/in^2 (minimum) 68,948 lbf/in^2 (minimum)
Tear strength, initial (Graves) MD XD	10.582 oz/mil (minimum) 10.582 oz/mil (minimum)
Tear strength, propagated (Elemendorf) MD XD	0.882 oz (minimum) 0.882 oz (minimum)
Dimensional change (Shrinkage) MD XD	3 percent (maximum) 3 percent (maximum)
Cold Brittleness	lower than -53.8°Celsius (C)
Water vapor transmission rate (WVTR)	0.882 oz/100 in^2/24 hrs (maximum)
MD = Machine direction XD = Transverse direction	

3.4 Workmanship. The workmanship shall be such as to ensure a product which is uniform and as specified herein. The film shall be clean, well finished and free from dirt, oil, foreign matter, rough or sharp edges, scratches, scuffs, cracks, creases, blisters, bubbles, pimples, pits, tears, cuts and holes, which might impair its usefulness for the intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test methods documents, and as follows:

a. Temperature, room ambient: 16 to 32°C

b. Altitude, facility ground

c. Humidity, facility ambient up to 95 percent relative humidity.

4.3.1 Inspection equipment. Unless otherwise specified, inspection equipment for conducting examinations and tests shall be as specified in applicable test method documents and paragraphs in this specification.

4.3.2 Lot formation. A lot shall consist of all material manufactured in one continuous operation using the same process, by the same manufacturer in accordance with this specification and submitted for inspection at one time.

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4.4 First article inspection. When specified in the contract or order (see 6.2), a first article sample consisting of four square feet of polyamide film manufactured under the same conditions as those proposed for subsequent production shall be subjected to first article inspection in accordance with table II. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained (see 3.1 and 4.2).

TABLE II. First article inspection.

Inspection	Requirement	Test paragraph
Specific gravity	table I	4.7.2
Tensile strength	table I	4.7.3
Tear strength, initial	table I	4.7.4
Tear strength, propagated	table I	4.7.5
Dimensional change	table I	4.7.6
Cold brittleness	table I	4.7.7
WVTR	table I	4.7.8

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table III (see 4.2).

TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Composition	3.2.1	4.7.1
Thickness	3.2.2	4.7.9
Workmanship	3.4	4.8
Packaging	5.1	4.9

4.5.1 Sampling. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities

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presented for inspection. Inspection sampling shall conform to table IV, or as specified herein.

TABLE IV. Sampling plan A.

Lot Size	Sample Size
2 to 3	100%
4 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29
NOTE: Acceptance Number is "ZERO" Reject Number is "ONE"	

4.6 Test methods and procedures.

4.6.1 Test sequence. Test sequence within each classification of inspection (first article or quality conformance) shall be at the option of the supplier.

4.7 Tests.

4.7.1 Composition. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the supplier shall, if specified in the contract or order (see 6.2), certify that the composition is in conformance to 3.2.1.

4.7.2 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 1505 to determine conformance to table I (see 3.3).

4.7.3 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 882 to determine conformance to table I (see 3.3).

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4.7.4 Tear strength, initial. The initial tear resistance shall be determined in accordance with ASTM D 1004 to determine conformance to table I (see 3.3).

4.7.5 Tear strength, propagated. The propagated tear resistance shall be determined in accordance with ASTM D 1922 to determine conformance to table I (see 3.3).

4.7.6 Dimensional change. The percent dimensional change shall be determined in accordance with ASTM D 1204 after a 30 minute exposure in an air circulating oven at 149°C to determine conformance to table I (see 3.3).

4.7.7 Cold brittleness. The cold brittleness temperature shall be determined in accordance with ASTM D 1790 to determine conformance to table I (see 3.3).

4.7.8 Water vapor transmission rate. The WVTR shall be determined in accordance with ASTM E 96 and to determine conformance to table I (see 3.3).

4.7.9 Thickness. Thickness shall be determined as specified in the contract or order to determine conformance to 3.2.2.

4.8 Examination. The film material shall be visually examined to determine conformance to requirements specified in 3.4.

4.9 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be as specified in the contract or order and MIL-STD-2073-1 and MIL-STD-2073-2 (see 6.2).

5. PACKAGING

5.1 Preservation, packaging and packing. Preservation, packaging and packing requirements shall be Level A, B, or C in accordance with MIL-STD-2073-1 and MIL-STD-2073-2 codes.

5.1.1 Level A. Level A shall be in accordance with the following codes:

MP	CD	PM	WP	CD	CT	UC	IC	PK	SM
2M	1	00	00	00	0,	XX	XX	E	99

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5.1.2 Level B. Level B shall be in accordance with the following codes:

MP	CD	PM	WP	CD	CT	UC	IC	PK	SM
2D	1	00	00	00	0	D3	XX	A	99

5.1.3 Level C. Level C shall be in accordance with the following codes:

MP	CD	PM	WP	CD	CT	UC	IC	PK	SM
10	1	00	YY	YY	X	YY	XX	L	99

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The polamide film material as specified herein is intended for use in the construction of a moisture barrier fly-through diaphragm in a missile container to provide protection against corrosion and deterioration prior to missile launch.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. When first article is required (see 3.1 and 4.4)

d. When certification of product is required (see 4.7.1)

e. Levels of preservation, packaging, packing and marking (see 4.9 and section 5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for

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examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (key word) listing.

Cellulose acetate sheet
Elastomer sheet
Nylon-6-type
Thermoplastic sheet

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 8135-A628

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-47249A (MI)	2. DOCUMENT DATE (YYMMDD) 920629

3. DOCUMENT TITLE
FILM, POLYAMIDE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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